Web Dev Workshop

Part II

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Recap

1. **HTML**: Markup language for structuring content.
2. **CSS**: Styling language for designing web pages.
3. **Tailwind CSS**: Utility-first framework for rapid and consistent styling.
4. **JavaScript**: Adds interactivity and dynamic behavior.
5. **Single File / Single Page Approach**: Built static websites using one file to encapsulate all content.

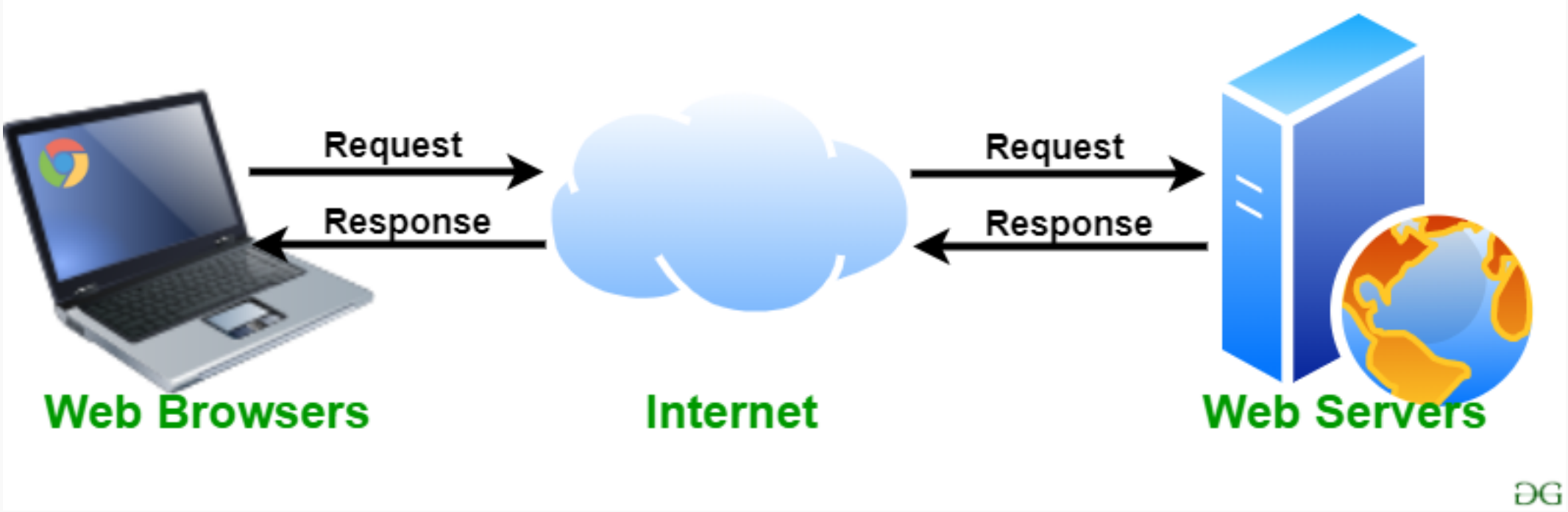
These fundamentals laid the groundwork for structuring website content.

Understanding Web Servers & HTTP Requests 🛜

Web Servers

1. A computer that stores website files.
2. Sends files to your browser when you visit a site.
3. Think of it as a waiter serving your order.

Web Servers


Request
Response
Request
Response
Web Browsers
Internet
Web Servers

What's an HTTP request?

1. It's a message your browser sends to a server.
2. It asks for specific content (like a webpage, image, or data).
3. The request includes details such as the method (e.g., "GET" for retrieving information) and the target resource.

In short, an HTTP request is the way your browser communicates with a server to get the information you need.

`https://www.google.com`

`https://www.youtube.com/watch?v=dQw4w9WgXcQ`

What’s an HTTP Response?

1. It’s the message your server sends back to your browser.
2. It includes a **status code** (e.g., 200 OK for success, 404 Not Found for errors).
3. The **body** carries the requested content (HTML, JSON, images, etc.).

An HTTP response tells you whether your request succeeded and delivers the content you asked for.

What is a Framework?

What is a Framework?

Definition

1. A framework is a set of pre-written tools and libraries.
2. It helps you build web applications faster.

Key Benefits

1. Organizes your code.
2. Offers reusable components.
3. Handles common tasks like routing and templating.

Frameworks let you focus on the unique parts of your app.

Exploring Different Frameworks

Frameworks Across Languages

1. Many frameworks exist for different programming languages.
2. JavaScript: React, Vue, Angular, Svelte.
3. Python: Django, Flask.
4. Ruby: Rails.
5. PHP: Laravel, Symfony.

Astro

In this workshop, we'll use Astro to build our project.

Our Choice: Astro

1. A modern framework for building fast websites.
2. Simple, efficient, and component-based.

Components

What are Components?

1. Reusable, self-contained pieces of UI.
2. Encapsulate structure, style, and behavior.
3. Allow you to build and maintain consistent interfaces easily.

Using Components in Astro

1. **Custom Components:** Build your own tailored to your needs.
2. **External Libraries:** Use ready-made components (e.g., shadcn) to speed up development.

Components help streamline your workflow and keep your code organized.

Let’s code 

Dependencies

Let's install VSCode and NodeJS

VSCode: `https://code.visualstudio.com/download`

NodeJS: `https://nodejs.org/en/download`

Code for this session

You can find the code for this session in the following link:

https://tinyurl.com/ieee-code

You can find the slides of this presentation here:

https://tinyurl.com/ieee-slideshow-presentation

Run the project

Install all the project dependencies

```
npm install
```

To run the project first initialize the shadcn components

```
npx shadcn@latest init
```

Then let’s run our web server

```
npm run dev
```

Exercises

Exercise 1

- Try and change all of the content tailored for you (hint: the files you want to change are inside the `/src/components`)
- Change the links to your own (email, linkedin, etc)

Exercise 2

- Inside the `/src/components/Experience.astro` there's a lot of repetitive code. How can you use JavaScript to automate the generation of code dynamically.

Exercise 3

- Now that you can grasp the codebase, try and create your own component and add it to your project. Feel free to adventure with ChatGPT and V0.

Solution

You can find my implementation in the following link:

`https://github.com/danielsp45/ieee`